

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

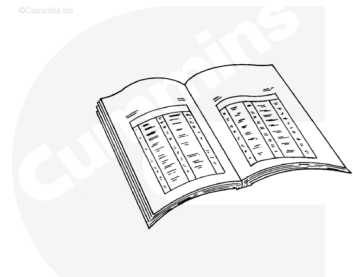
⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a lift or get assistance to lift this component.

- Drain the lubricating oil and remove the lubricating oil pan. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>).
- Remove the suction tube and block stiffener. Refer to Procedure 001-089 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).

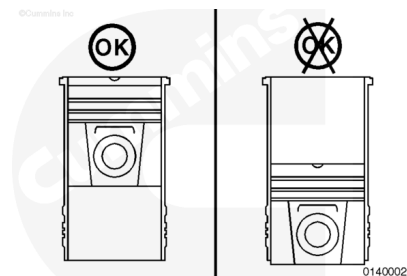


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Remove

⚠ CAUTION ⚠

Do not attempt to remove the piston cooling nozzle when the piston is at bottom dead center. The nozzle will be damaged and can result in piston damage.

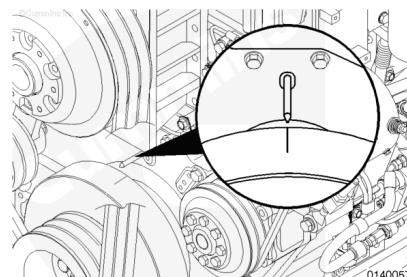


To reduce the possibility of damage to the piston cooling nozzles during removal, use the valve set marks and the sequence chart to identify those nozzles that can be removed at each index mark on the damper.

One complete crankshaft revolution is required to remove all of the nozzles.

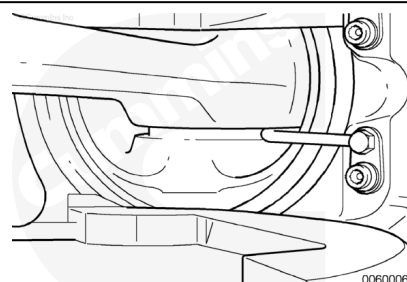
Piston / Position	Remove Nozzles
Number 1 TDC	1 and 6
Number 2 TDC	2 and 5
Number 3 TDC	3 and 4

Rotate the engine barring device until the "1,6" mark on the damper is aligned with the front cover pointer.

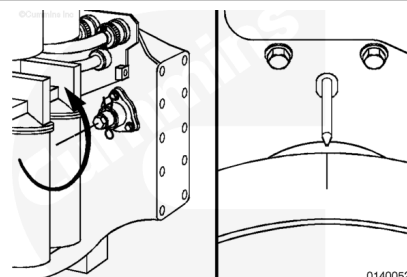


Remove the piston cooling nozzle capscrew.

Remove the front and rear piston cooling nozzles on cylinders number 1 and 6.



After removing the nozzles from cylinders 1 and 6, rotate the engine barring device and align the next valve set mark on the damper with the pointer on the front cover.



Use the chart in the illustration to determine the appropriate cooling nozzles to be removed.

Repeat the process to remove the remaining piston cooling nozzles.

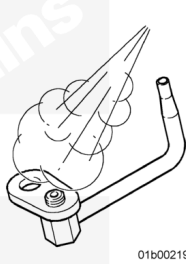
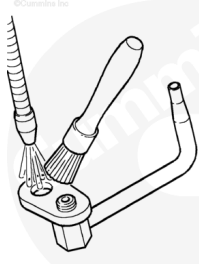
Piston / Position	Remove Nozzles
Number 1 TDC	1 and 6
Number 2 TDC	2 and 5
Number 3 TDC	3 and 4

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Clean

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

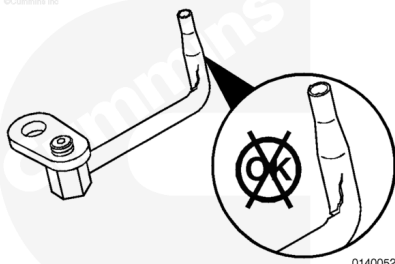
Use solvent to clean the piston cooling nozzles. Dry with compressed air.

Blow out the oil passages with compressed air.

Inspect for Reuse

⚠ CAUTION ⚠

Any damage to the piston cooling nozzle can result in major engine damage.



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Check the nozzle for cracks, bends, or other damage. Check the spray hole for burrs.

The piston cooling nozzle **must** be replaced if it is damaged.

Install

⚠ CAUTION ⚠

Do not attempt to install the piston cooling nozzle when the piston is at bottom dead center. The nozzle will be damaged and can result in piston failure.

Piston / Position	Remove Nozzles
Number 1 TDC	1 and 6
Number 2 TDC	2 and 5
Number 3 TDC	3 and 4

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Rotate the engine barring device to position the cylinders for best access to the piston cooling nozzle mounting pad.

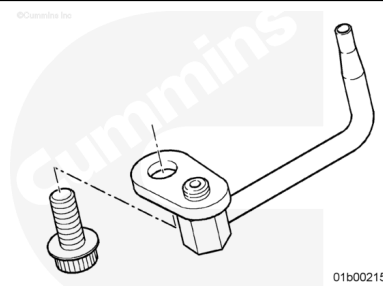
Refer to the sequence chart in the illustration to identify the four piston cooling nozzles that can be installed at each timing mark.

Install the piston cooling nozzles stamped with an “F” at the front of the two cylinders selected.

Install the piston cooling nozzles stamped with an “R” at the rear of the two cylinders selected.

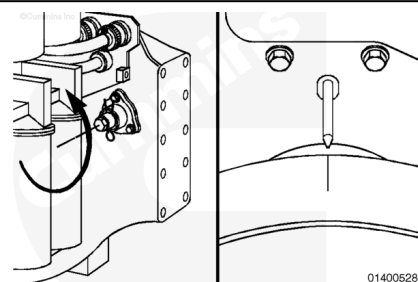
Tighten the capscrews.

Torque Value: 30 N·m [22 ft·lb]



Rotate the barring device and align the next valve set mark on the damper with the pointer on the front cover.

Repeat the process to install the remaining piston cooling nozzles.



Finishing Steps

- Install the suction tube and block stiffener. Refer to Procedure 001-089 (</qs3/pubsys2/xml/en/procedures/89/89-001-089.html>).
- Install the lubricating oil pan. Refer to Procedure 007-025 (</qs3/pubsys2/xml/en/procedures/89/89-007-025.html>).

- Fill the engine with clean 15W-40 lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>) for engine oil capacity.



Last Modified: 30-May-2003
